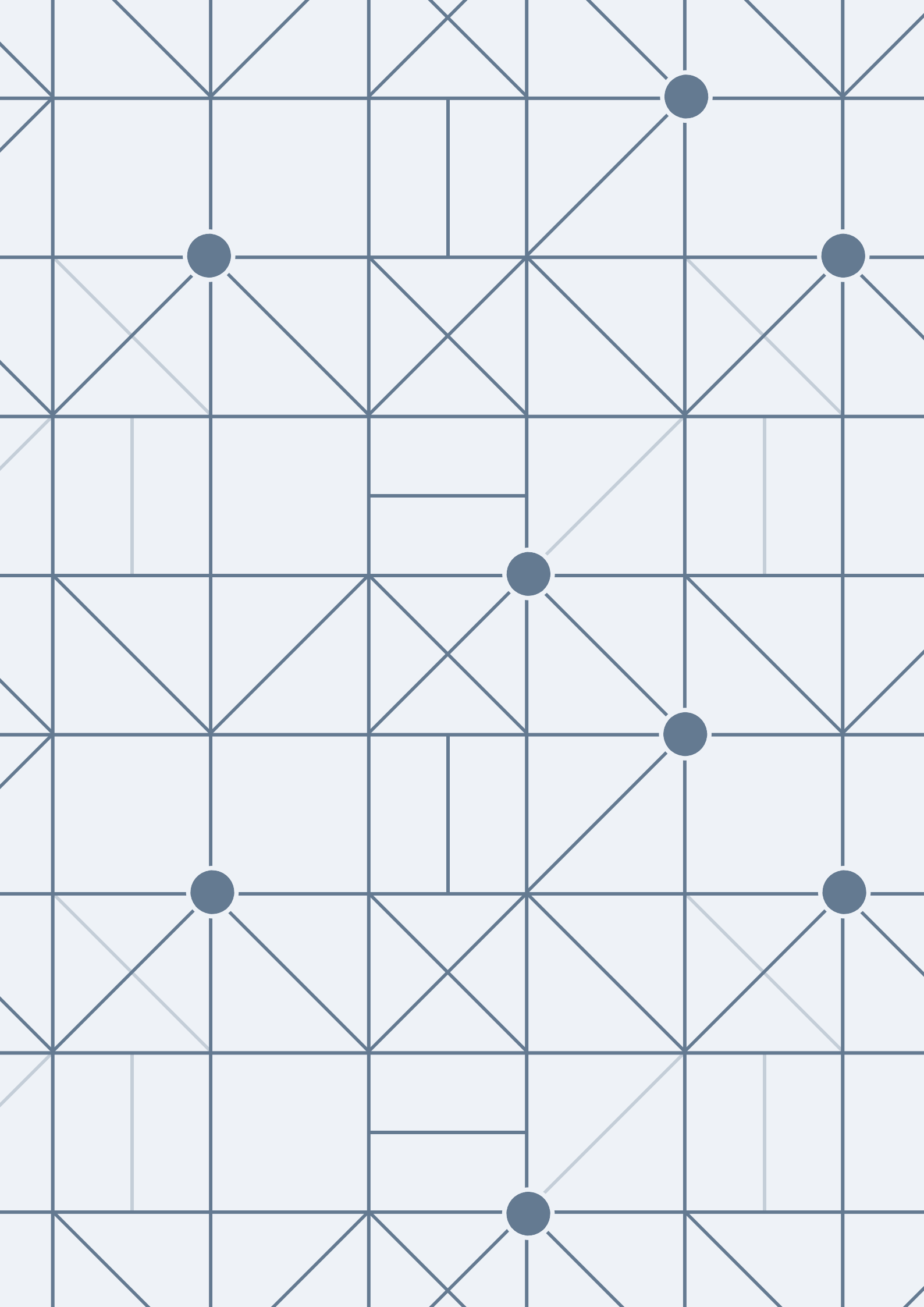


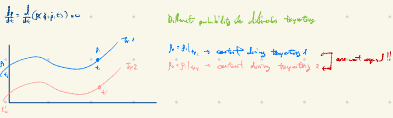
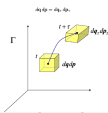


Liouville's Theorem

The density of an ensemble of systems of thermodynamic equilibrium remains constant as they evolve according to Hamilton's equations of motion

$$\frac{\partial \rho}{\partial t} + \{\rho, H\} = 0 \Rightarrow \frac{d\rho}{dt} = 0$$

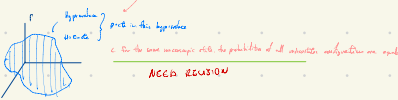
$$\{\rho, H\} = 0$$



$$\frac{d\rho}{dt} = \frac{\partial \rho}{\partial t} + \frac{\partial \rho}{\partial q} \dot{q} + \frac{\partial \rho}{\partial p} \dot{p} = 0$$

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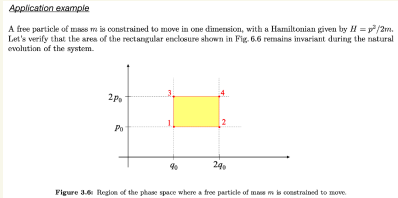


Liouville's Theorem

$$\{\rho, H\} = 0 \Rightarrow \rho = \text{const}$$

$$\rho \sim e^{-E/KT}$$

$$\rho \sim e^{-E/KT + \mu N/KT}$$



$$\alpha(q_0, p) \rightarrow \alpha(q_0 + \Delta q, p)$$

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$$\frac{d}{dt} \left(\frac{p}{m} \right) = 0 \Rightarrow p = mv$$

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$$d\rho = \frac{\partial \rho}{\partial q} dq + \frac{\partial \rho}{\partial p} dp$$

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